SQL Views Practice Assignment

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Schema:

CREATE TABLE students (
    student_id INT PRIMARY KEY,
    name VARCHAR(50),
    dept VARCHAR(50),
    marks INT
);

INSERT INTO students VALUES
(1, 'Aman', 'Computer', 85),
(2, 'Riya', 'Mechanical', 78),
(3, 'Kabir', 'Computer', 90),
(4, 'Sneha', 'Civil', 65),
(5, 'Arjun', 'Mechanical', 72);

CREATE TABLE courses (
    course_id INT PRIMARY KEY,
    course_name VARCHAR(50)
);

INSERT INTO courses VALUES
(1, 'DBMS'),
(2, 'Thermodynamics'),
(3, 'Structures');

CREATE TABLE enrollments (
    student_id INT,
    course_id INT
);

INSERT INTO enrollments VALUES
(1,1),
(2,2),
(3,1),
(4,3),
(5,2);
```

Questions:

Create a view to show student name and marks.

Create a view to show students with marks greater than 80.

Create a view to show student name and course name.

Create a view to show average marks.

Create a view to show department wise student count.

Create a view to show students enrolled in DBMS.

Create a view to show student name and number of courses enrolled.

Create a view to show top 2 students based on marks.

Create a view to show students from Computer department.

Create a view to show course name and number of students enrolled.

Create a view to show students who scored above average marks.

Create a view to show only students with marks greater than 70 using check option.

Create a nested view (one view based on another).

Try updating data using a view and observe.

Explain what happens when base table data changes.